

# Aleksander Borodin

ML Engineer | LLMs & AI Agents

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## Summary

ML engineer and Applied Mathematics and Physics student at MIPT. I build LLM systems, from RAG and model integrations to multi-agent workflows and agent-driven evolution: iterative search and improvement of solutions. My experience includes benchmark development, model evaluation, and applied optimization.

## Education

### Bachelor's degree, Applied Mathematics and Physics

2023–2027

FPMI MIPT, in progress; GPA 7.8/10

Coursework: mathematical analysis, probability, linear algebra, algorithms, computational complexity, ML, physics, economics.

## Professional Experience

### ML Engineer - Research Internship

Feb–Aug 2026 (6 months)

CKM, MIPT

- Developed LLM-based agents and benchmarks to evaluate their performance.
- Worked on agent-driven evolution: systems in which agents iteratively generate, evaluate, and improve solutions.
- Contributed to two research papers: one under review at ACL and another in preparation for submission.

### Product / AI Engineer

Jan–May 2026

ModelGate.ru – LLM API gateway and proxy service

- Developed an LLM API gateway providing a unified interface to models from multiple providers.
- Integrated provider APIs and implemented request routing with model and provider selection.
- Built core service components, including the backend and public website. Stack: Python, LLM APIs.

### Model and Algorithm Developer

Sep 2024–Feb 2025

Regional Bank, client project

- Developed a cash collection optimization algorithm; historical-data analysis projected a 20M RUB annual profit increase.
- Built a CBR web-scraping/analysis pipeline for data unavailable via API and an internal document Q&A RAG system.
- Stack: Python, NumPy, Pandas, RAG, web scraping.

## Projects

### Idea Evolve

2026

github.com/aleksanderborodin/idea\_evolve

- Evolutionary multi-agent framework: specialized Claude agents work in parallel across generations and build a shared knowledge base.
- Improved Sidon set search from 66 to 105 elements in 7 autonomous generations, passing the  $\sqrt{N} \approx 100$  reference for  $N = 10000$ .
- Implemented role-specialized agents, resumable filesystem state, evaluation/knowledge extraction, and a Flask dashboard.
- Stack: Python 3.12+, Claude Code CLI, OpenCode, Anthropic API, Flask, multi-agent orchestration.

### Prefix Injection Jailbreak

2025

github.com/aleksanderborodin/prefix\_injection\_jailbreak

- Researched Choice-Blindness-inspired prefix attacks; templates made Gemini 2.5 Pro and DeepSeek R1 complete MaliciousInstruct prompts with near-100% success vs. a near-zero baseline.
- Stack: Python, LLM APIs, model evaluation, prompt/context engineering.

### Yoola Explain

2025

github.com/aleksanderborodin/yoola\_explain

- Browser extension that extracts Terms of Service and returns structured LLM summaries: key points, data collection notes, and warnings.
- Added multi-language support and caching. Stack: JavaScript, FastAPI, OpenRouter, LLM APIs.

## Skills

**Programming** Python advanced; C++ intermediate; C and Assembler basic code reading.

**Data / ML** PyTorch basic; NumPy; Pandas; PostgreSQL; Matplotlib; ClearML; Excel.

**NLP / LLM** BM25; embeddings; RAG; LLM APIs; evaluation; prompt/context engineering; jailbreak testing.

**Agentic tools** Claude Code CLI; OpenCode; multi-agent orchestration; filesystem-based workflows.

**Backend** FastAPI; Flask; LangChain; LangGraph; Linux home server administration and workstation usage.

**Languages** Russian native; English B2+ certified by MIPT placement test.

## Programs & Achievements

### Summer with AIRI 2026 - Artificial Intelligence Summer School

Jul 21–Aug 4, 2026

Final team project: Ideas Lake – long-term memory for evolutionary AI workflows using hybrid retrieval and a knowledge graph.

**Gold Medal**, International Experimental Physics Olympiad (IEPhO)

2023

**Prize-winner**, All-Russian School Olympiad in Physics, final stage — Grades 10 and 11

2022, 2023

Winner, team Wiki-races, Sirius

2023